

Lab 7.5 Analyzing a Research Methods study

This lab is designed to show you how this data analysis would work in research methods. In this lab, you will analyze some questions based on a past group's research study. Part of this lab is based on your own ideas given the study here.

The group that did this study was interested in how parents impact a person's happiness, self-efficacy, and belongingness.

Set up an R notebook and download the data, which is named "lab7_5.csv". This is not a normal CSV file but one exported from Qualtrics.

Step 1 The first thing you should do is load the tidyverse, psych, and qualtrics libraries. You will be loading data exported from Qualtrics.

Step 2 Import the data using the readSurvey command from the qualtrics package. If you have issues here, look at Lab 2 which talks about importing qualtrics data.

The data has 57 observations of 128 variables. That's a lot! But fortunately we are only going to focus on a few variables. These are all the way on the right of the data set and you may have to scroll over and push the » that's beside "cols" on the top of the data viewer to see the variables of interest.

Here are the variables that we're interested in:

- Q2 - gender (female or male)
- SC0 - a measure of self-efficacy
- SC1 - a measure of happiness, with low scores indicating less happiness, more depression
- SC2 - a measure of how much a person feels like they belong
- SC3 to SC8 - measures of a person's parental impact, or how much a person thinks their parents impacted them.
- SC3 - how available parents seemed
- SC4 - how supportive parents seemed
- SC5 - how understanding parents seemed
- SC6 - how respectful parents seemed
- SC7 - how restrictive parents were
- SC8 - how much guidance parents seemed to give

Question 1: Note all of these are self-report measures that students answered about their parents. What are some problems with this approach when we are trying to measure how parents actually acted?

Step 3 The next step will be to remove any bad data. In this case, they had some survey previews. Type in the following code chunk to remove survey previews. When you do a Qualtrics study, you should include this.

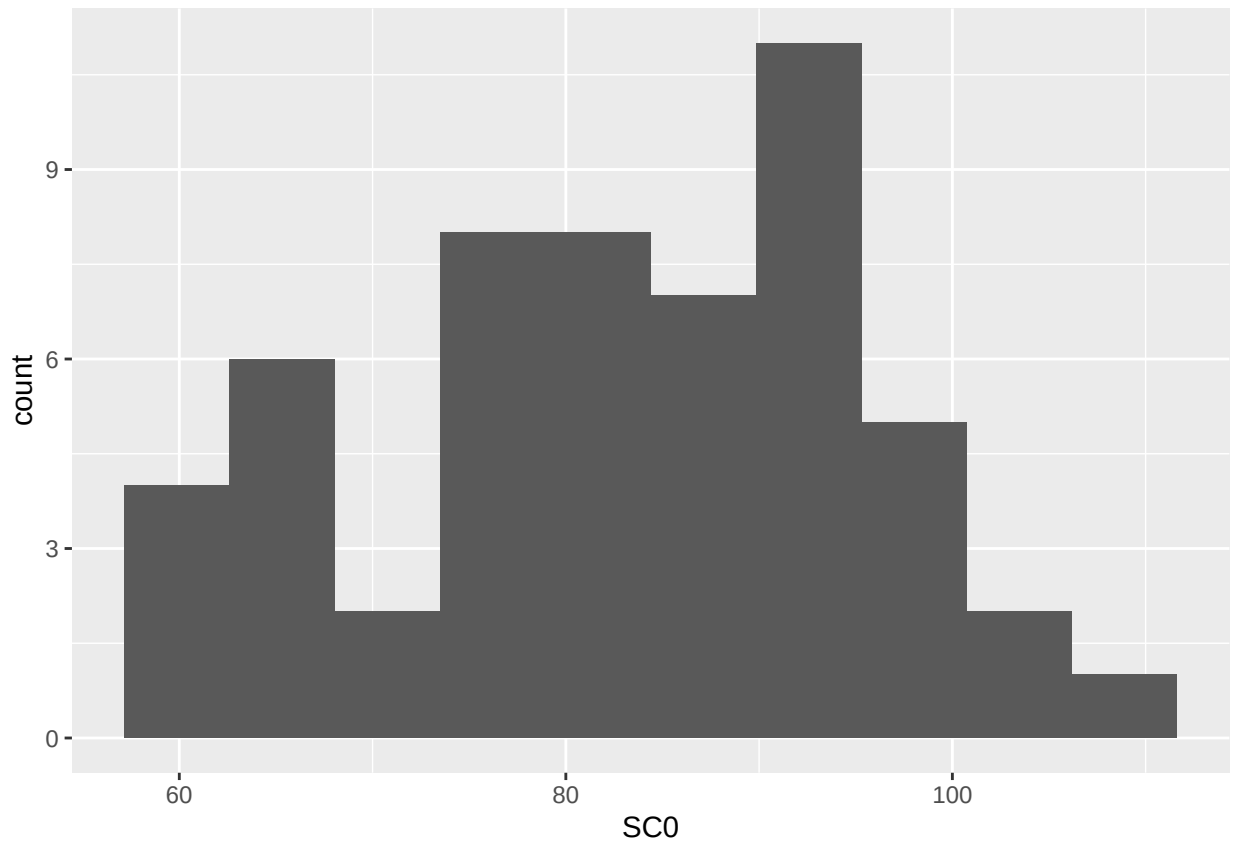
```
d = filter(d, Status != "Survey Preview")
```

Step 4: One of the first steps will be to see how many females and males completed the study. Type the following table() command to output the gender breakdown.

```
table(d$Q2)
```

Step 5: Use the `table()` command like above for variable Q3 which looks at parents' marital status. This should generate a table which lists how many people were in each condition of marital status.

Step 6: Create a histogram to view the SC0 variable and see if there are any outliers for self-efficacy. Use `ggplot` and 10 bins. It should look something like this graph below: Let me know if you think there are outliers:



Step 7: Use the `describe()` command to view summary statistics for each of the parental impact scales, which are SC3 to SC8.

Step 8: Use the `describeBy()` command to see whether it looks like self-efficacy varies by gender. You would type the following code to do this:

```
describeBy(d$SC0, d$Q2, mat=T)
```

Step 9: Conduct a t-test to see if there is a significant difference between females and males in self-efficacy. Report in your annotations whether this is significant and what the t-value and p-value are.

Step 10: A researcher might think that parental impact is different for girls than boys. Take two of the parental impact scales and see if they are significantly different for girls than boys, using a t-test. Remember, the parental impact scales are SC3 to SC8

Step 11: The researchers are interested whether there's a significant correlation between parental impact and self-esteem. Pick a parental impact variable you think is correlated to self-esteem and use the `cor.test()` function to see if there is a significant correlation.

Step 12: Use a scatterplot to visualize your correlation. Give it appropriate labels. The code below looks at the correlation between Supportive parental impact and Depression-happiness. You can use it as an example.

```
ggplot(data = d, aes(x = SC4, y = SC1)) + geom_point(color = 'dark green') +  
  geom_smooth(method = 'lm', se = F, color = 'red') +  
  labs( x = "Supportive", y = "Depression-Happiness", title = "the title")
```

Step 13: Last step! Look at the variables and figure out an association or group mean hypothesis and test it out. You should use either a correlation or a t-test to test your hypothesis. Do the appropriate analysis and write an explanation stating whether you think your hypothesis is supported or not, based on the data.

Now knit your notebook and upload the knitted version and you're done!